Training : GIT and GITHUB

Agenda :

DAY 1 agenda :

> Introduction to version control tool - done
> Git and github - done
> git vs github - done
> Benefits - done
> Creating a github account and repository
> local and remote repos
> repo structure
> readme and .gitignore - next session
> basic git workflow (init -> add -> commit)

we save the code locally as well as over the central repo .

Distributed version control system (DVCS)

How we save the versions locally?
How we maintain the version over centrol repo?

To save and maintain the code as version locally we use the tool : GIT (tool which is downloaded in your machine) developed by linux foundation

central repo :
github (owned microsoft)
gitlab
bitbucket
code commit (aws)

```
Administrator@EC2AMAZ-4F2NA2A UCRT64 ~/Desktop/project-folder (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.txt

nothing added to commit but untracked files present (use "git add" to track)

Administrator@EC2AMAZ-4F2NA2A UCRT64 ~/Desktop/project-folder (master)
$ >
```

git status will tell if git is tracking the files/ folders in your working repo or not
But what is tracking?
if git is tracking the content of your file then intentionally or unintentionally if you make any changes git will be able to identify it

Working repo

First step : I will start tracking the files ...means that if i make any changes to the content of the file ...git will immediately be able to tell me

git add filename or git add .

Second step : You need to tell the git who is saving the code as a version .

Third step :The content which git is tracking will be saved as a version ....

30 MIN TASK :

For those who have done it:

> create new file

test1.txt test2.txt test3.txt test4.txt

put some content in the file

> check the status

> now start tracking test2.txt and test3.txt

> now make changes in the content of file test4.txt and test3.txt with the content of your choice

> now check git status again

> can you see the difference at the content level

> now add start tracking test4.txt file

> later you reliase that you dont need to test2.txt to be saved as a version..and you need to move it back to non tracking file..how you can do it...

> now check which files git is tracking and which git is not tracking

> Lets suppose your name is Ronaldo with mail : ronaldo@ymail.com ..ensure that your versions should have that details

> save the code as a version and show the versions created

Day 2 Agenda:

1) Working with remote repos
2) Clone github repo
3) Push and pull changes
4) Sync local and remote repos
5) Intro to branches
6) Why branches required
7) Feature based development
8) Basic merge concepts
9) Introduction to merge conflicts

Handson exercise:
Clone an existing github repo
Create a feature branch and make changes
Merge a feature branch into main branch
Create and resolve a basic merge conflicts

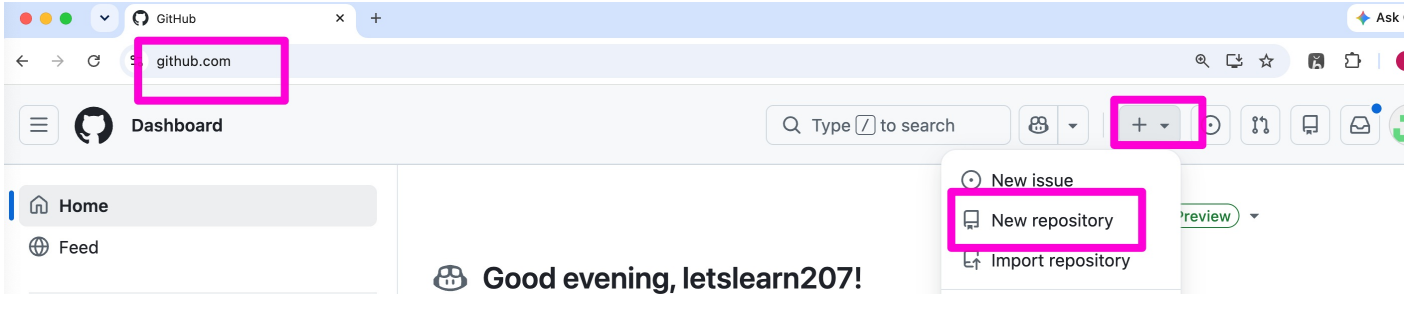
Revision : 1 day session
LAb:
Created a folder -> inside the folder -> opened the git bash
Created a file - index.txt in folder with content " hello world"

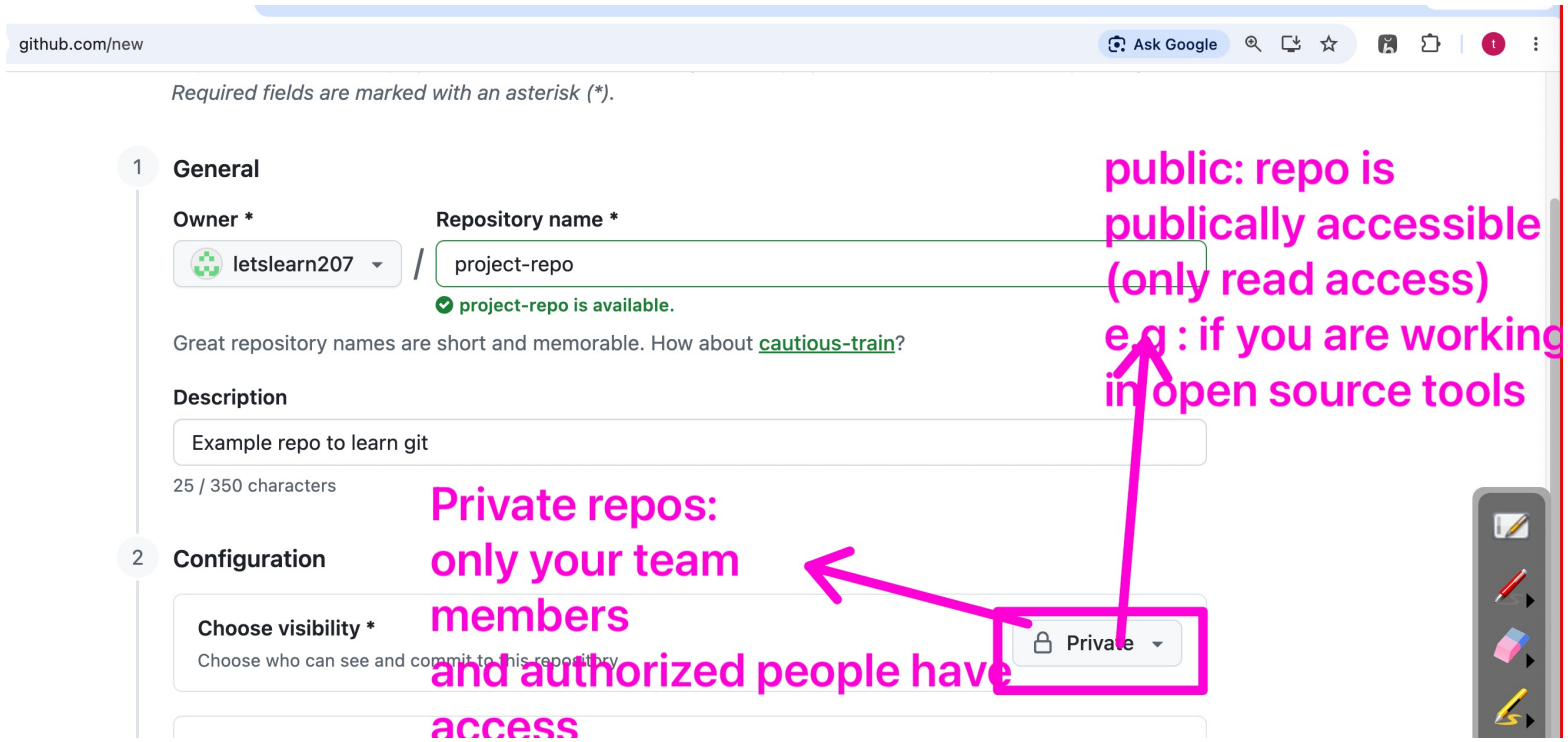
In bash,
git init # initialise the git
git status # tells if git is tracking the file or not
red color : git is not tracking the file
green color: git is tracking the file
#add command means start tracking the file
git add index.txt or git add . # here . means all the files
one time command : We need to tell /configure git that who is working
git config global --user.name "AKSHAT"
git config global --user.email "akshat@gmail.com"
save the code as a version:
git commit -m"my first commit"
to see the versions : --> git log
git log --oneline # to see the versions in single line

today we will work with remote repos :

go to github.com -> login

Click on create repo





No other changes

Create repo

development environment.

Create a codespace

Invite collaborators

Quick setup — if you've done this kind of thing before

Set up in GitHub Desktop or HTTPS SSH https://github.com/letslearn207/project-repo.git

Get started by creating a new file or uploading an existing file. We recommend every repository include a README, LICENSE, and .gitignore.

...or create a new repository on the command line

```
echo "# project-repo" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/letslearn207/project-repo.git
```

Remote repo

This is a blank remote repo

We have the versions locally (in our machine) which we need to push to this remote repo

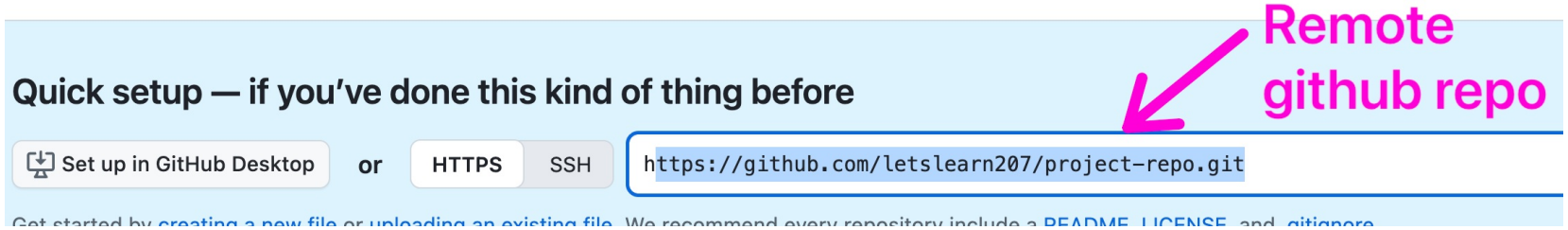
Go to novelvista labs :

go to the folder you started working on yesterday

Right click and open git bash

git log --oneline # check if you have versions avaiable

In my case :



so i need to tell my local that https://github.com/letslearn207/project-repo.git is my remote repo

In git bash,

git remote add origin https://github.com/letslearn207/project-repo.git

this means that i want to add the remote repo as a origin for my local ...means whenever i will the push the code it will automatically push it to https://github.com/letslearn207/project-repo.git and if i pull the code it will be pulled from https://github.com/letslearn207/project-repo.git...

```
akshat@Akshats-MacBook-Air git-cgi % git remote add origin https://github.com/letslea
rn207/project-repo.git
akshat@Akshats-MacBook-Air git-cgi % git remote -v
origin  https://github.com/letslearn207/project-repo.git (fetch)
origin  https://github.com/letslearn207/project-repo.git (push)
akshat@Akshats-MacBook-Air git-cgi %
```

git remote -v

-v : verbose : all the details
git remote : Remote repo which we have added

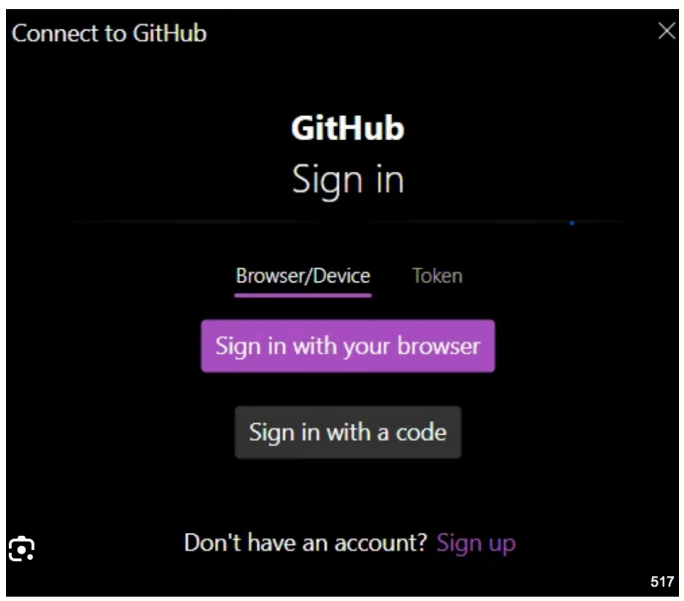
now since we have set the remote as origin

we will now push the code to github :

git push -u origin master

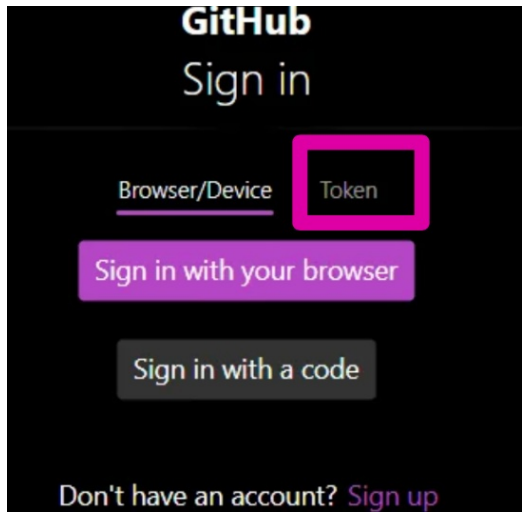
we want to push the code from local to github (upstream) to my remote repo (origin : github.com repo which we have set in previous step) to master branch

In windows ,
You will see a pop up appearing ....(if you dont find the pop look at the startup menu ...you will see a pop up)

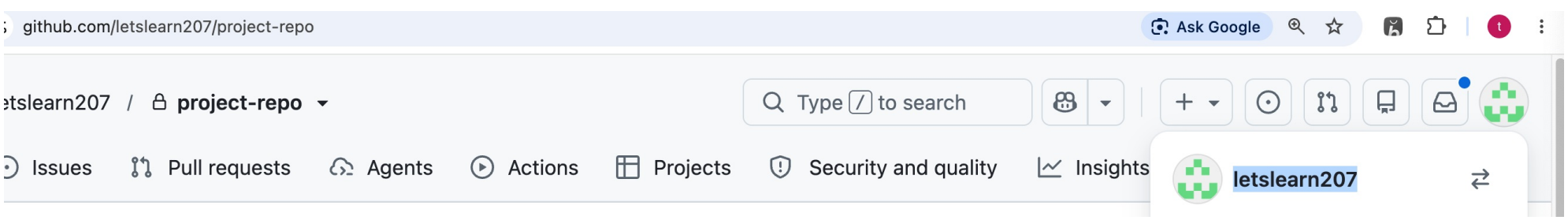


Click on token

#TOKEN : Authentication token which we will use to connect to github



If it ask for username: put the github username

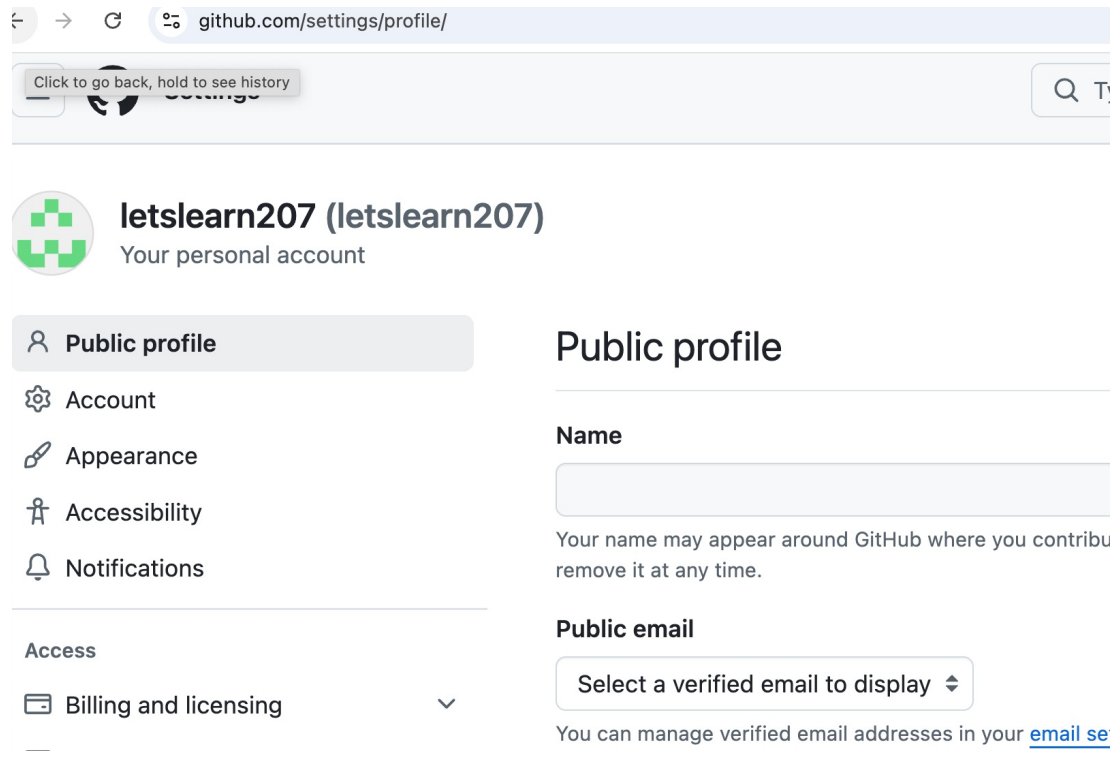


Lets now generate token as well

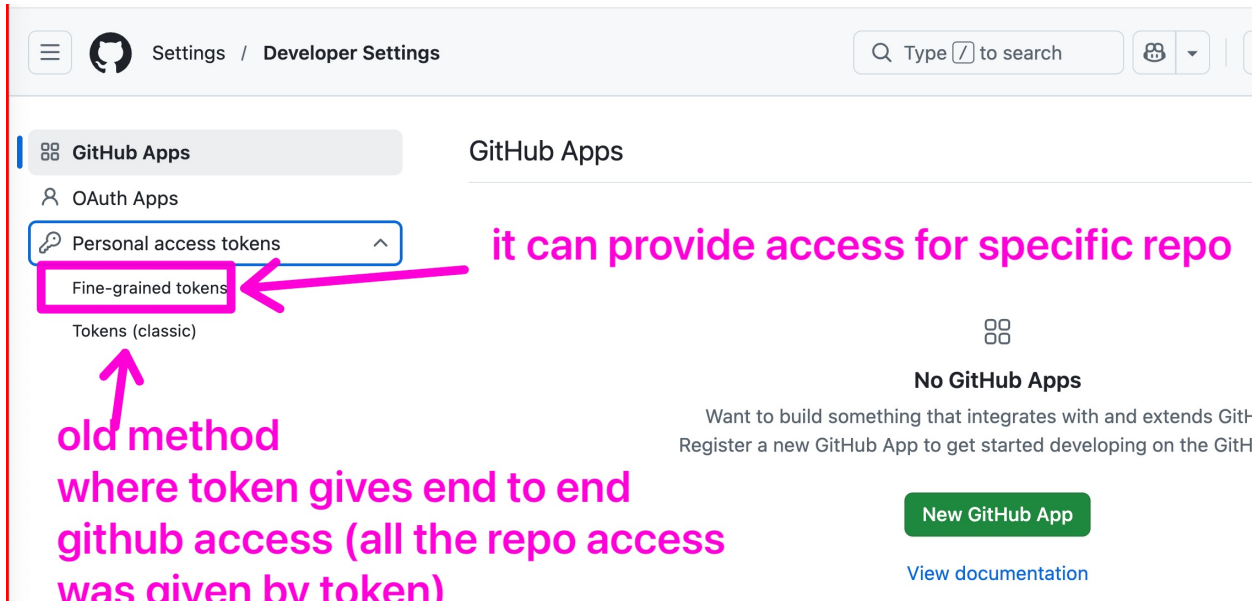
Go to github.com

On the right side you will see the profile (circle) 

Click it and open settings



Scroll down and find Developer settings
(or open this link https://github.com/settings/apps)



it can provide access for specific repo

old method
where token gives end to end
github access (all the repo access
was given by token)

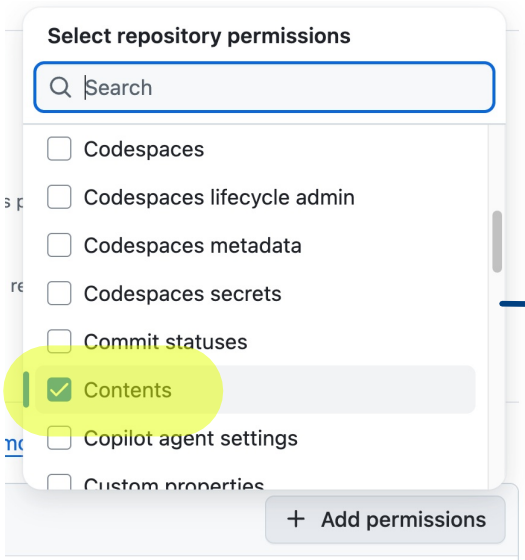
Click on fine grained token

> generate new token

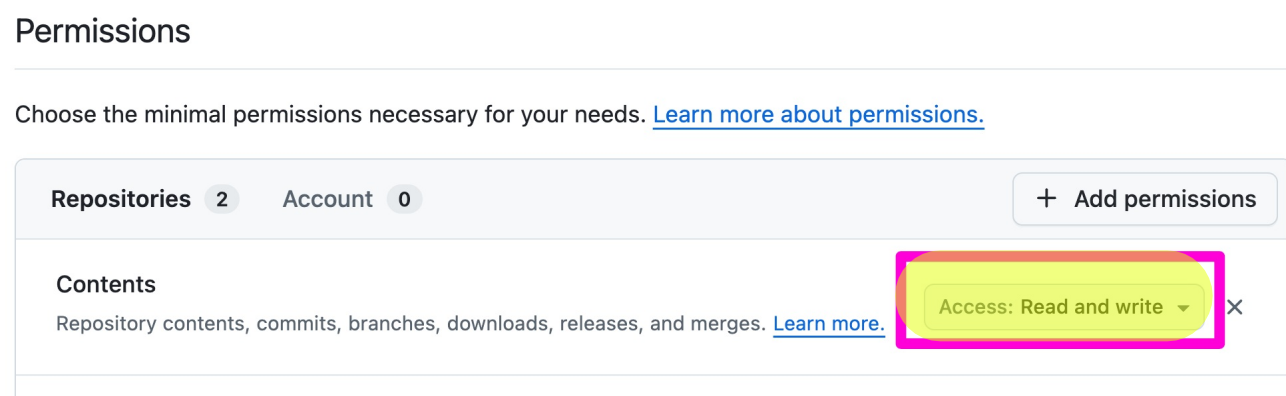
Give the token name
Expiry date: any date
Repo access: All repos

Select contents only

+ Add permissions



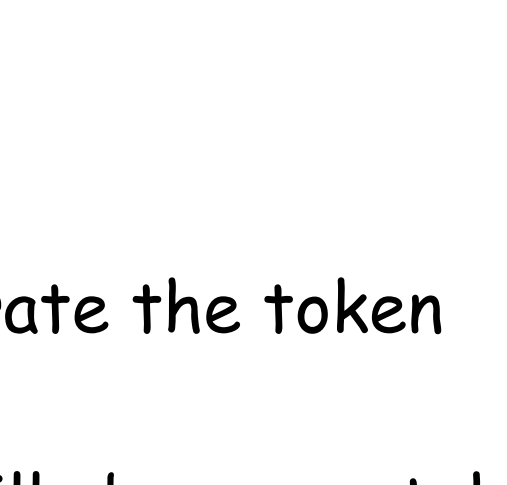
change the content to Read and write



Generate the token

you will observe a token ...copy and put it somewhere (like in notepad)

In local:



it will ask for token

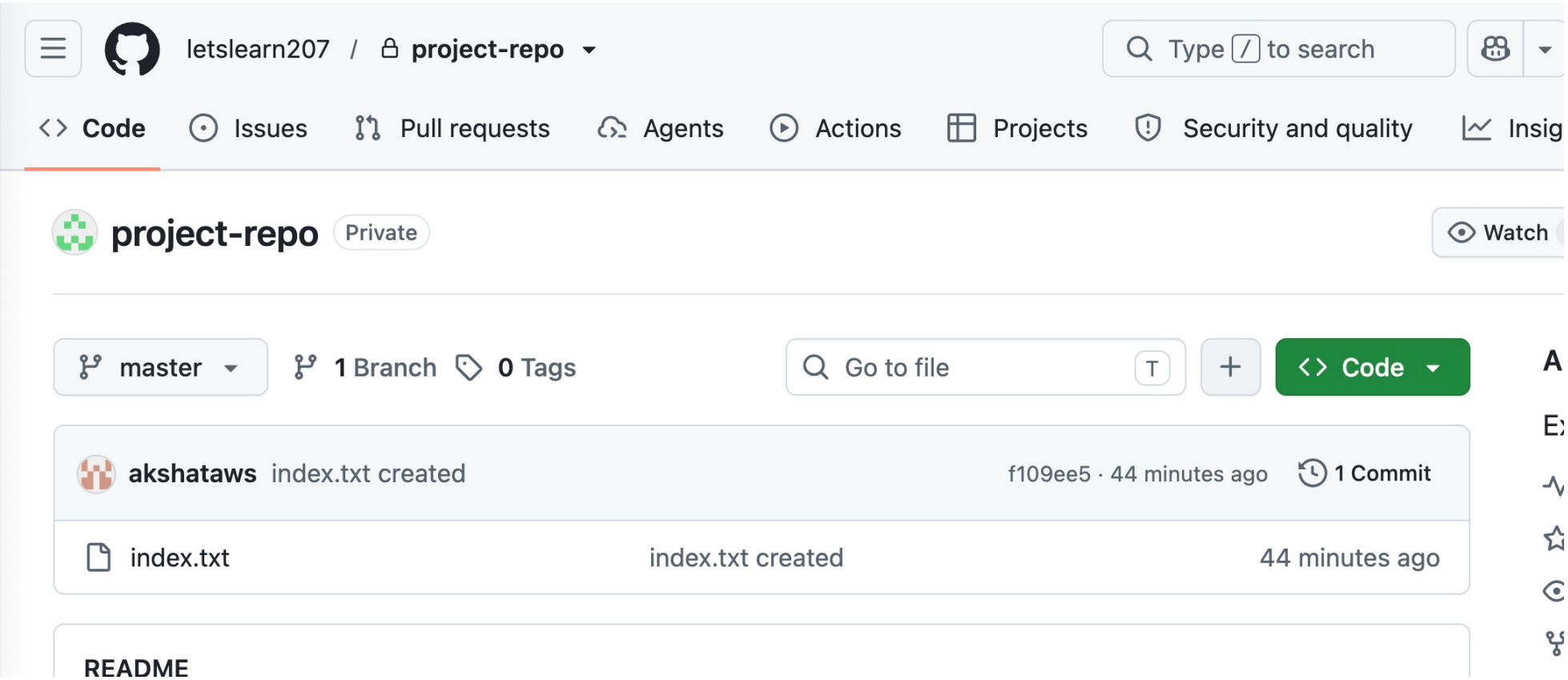
put the token which you have generated here (popup)

As soon as you put the token and click on ok
your code will pushed to the github ...
if you get an error here...check the token permissions and recreate the token

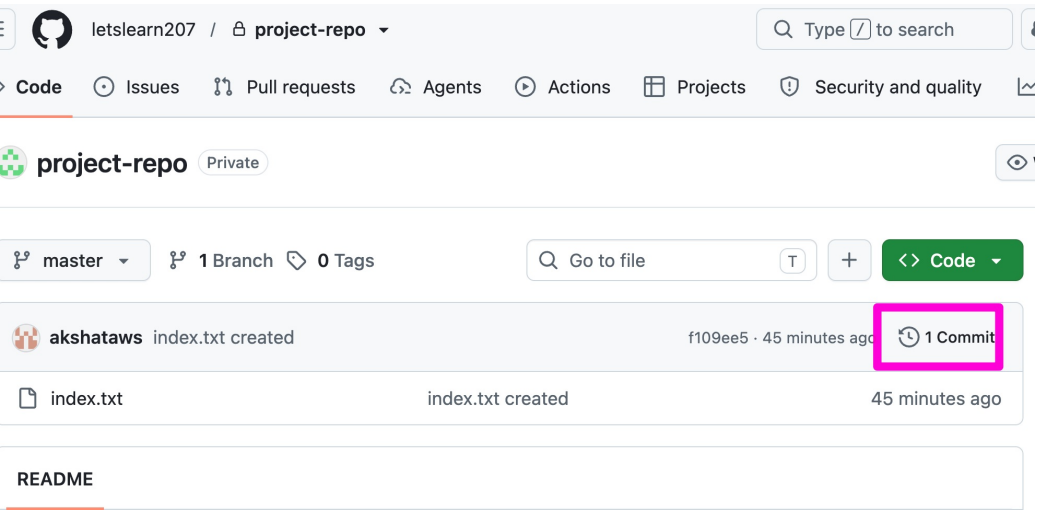
Lets now check if the files from local is actually pushed to github or not

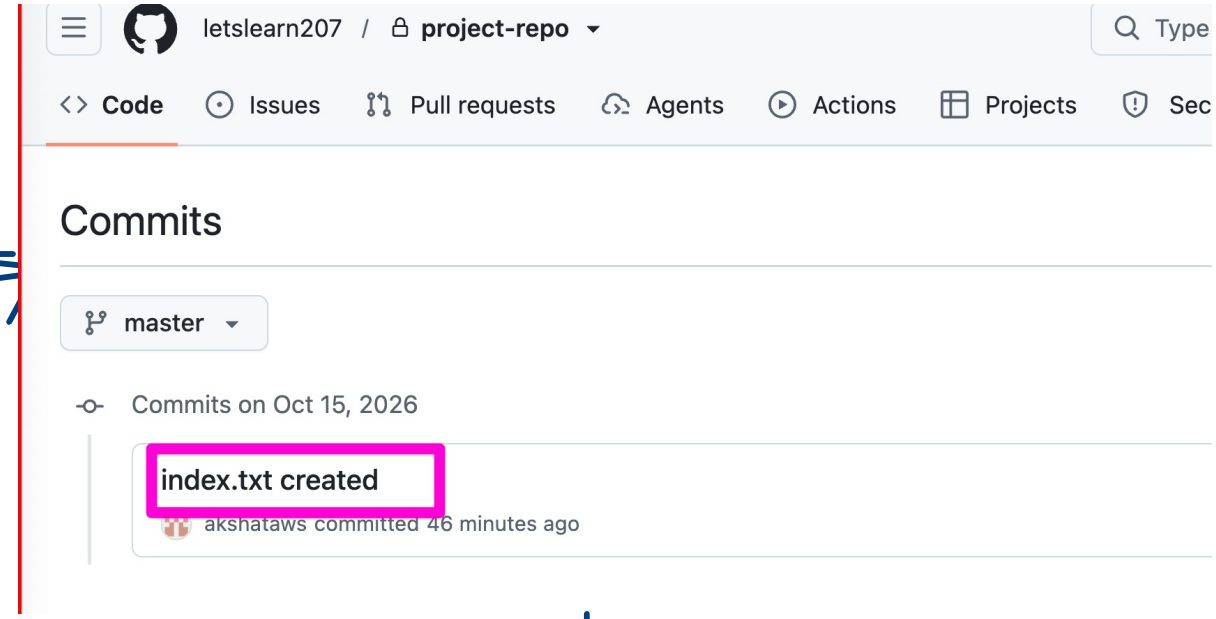
go to github.com and go to the repo

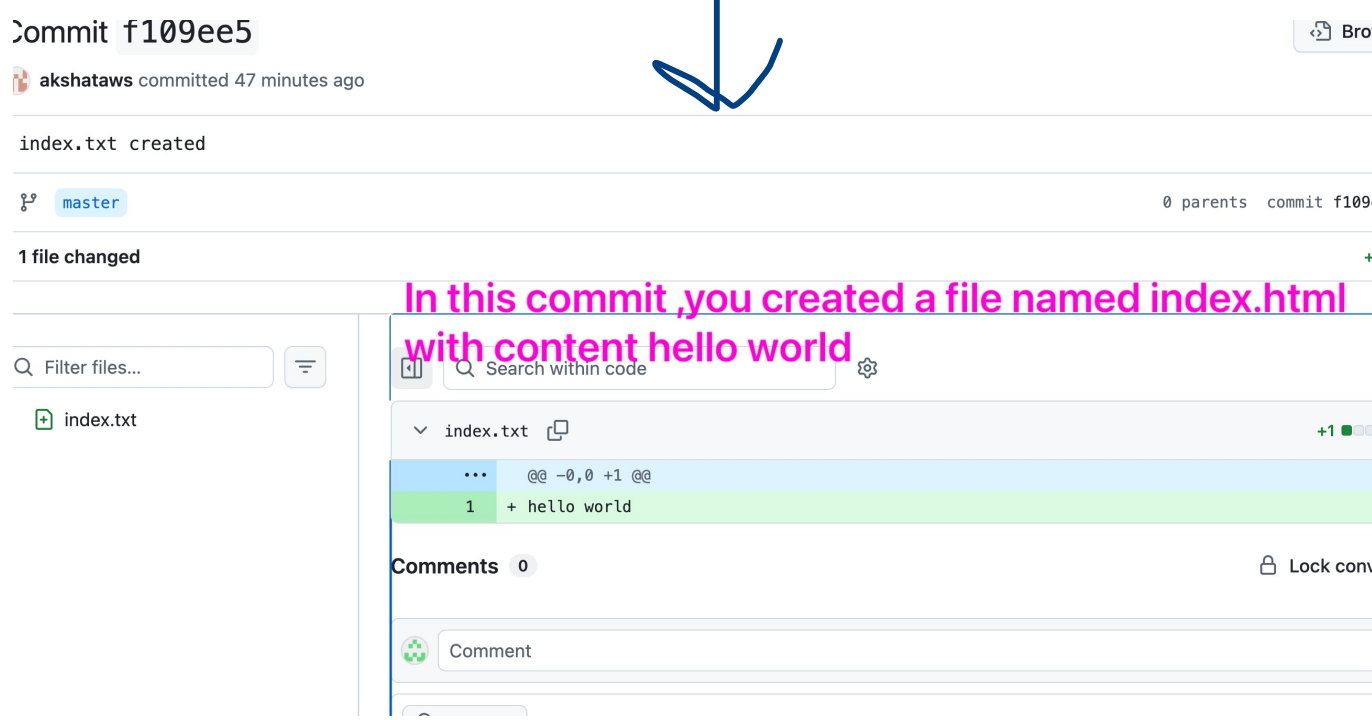
if you have forgotten the repo : in gitbash -> git remote -v



To identify what you have done in the specific commit







LAB 2

SUPPOSE YOU ARE JOINING A PROJECT which already has a repository ..

Take an example: You joined a team which is working on apache project

and already have this repo : https://github.com/letslearn207/apachewebsite

now you will be working on the existing repo ...you will not be creating a new repo like we did in lab 1

First of all go to novelvista lab or your laptop:

go to desktop -> right click and open gitbash

git clone https://github.com/letslearn207/apachewebsite

cd apachewebsite

now you will see a folder created in your novelvista lab /laptop with name apachewebsite

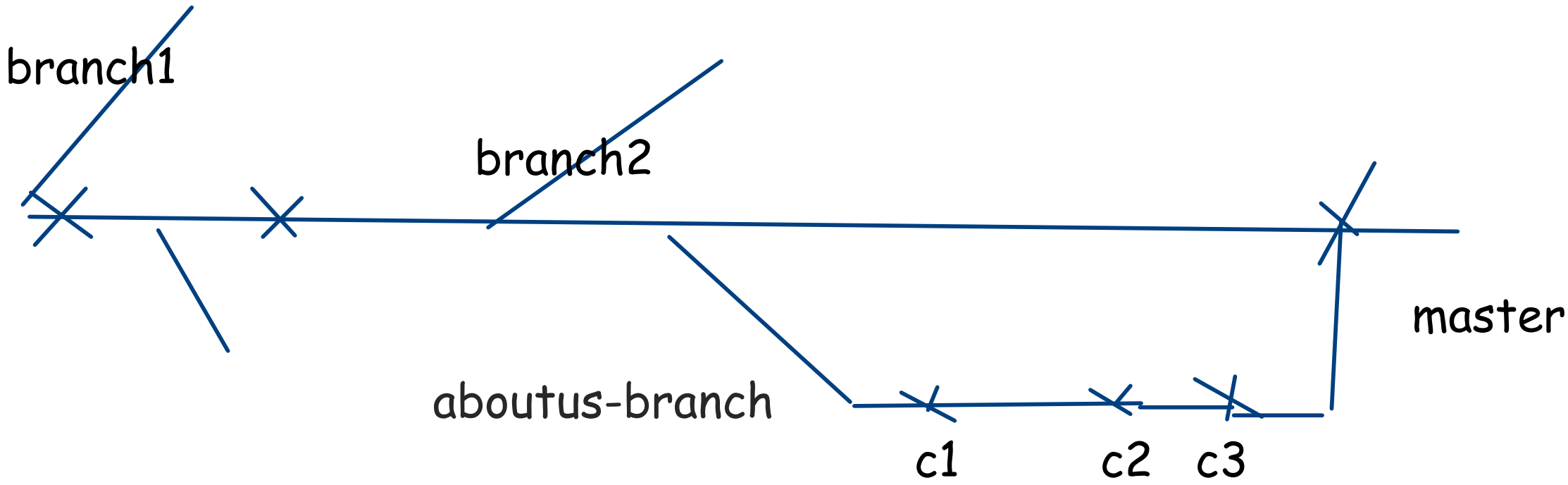
when you enter inside the folder you will observer all the files and commits done by other developers

now lets say you are a developer and assigned a task to design a webpage : aboutus page ...

As a developer you will always work in a branch and when you reliase that your functionality is developed you will merge the code in the master

Feature Branch: Provides an isolated environment where the developer can work

master/main: branch where all my developers code will come and final deployment will happen in master branch



you can see the list of branches

git branch
o/p * master or * main (* means current)

create a new branch :
git branch branchname

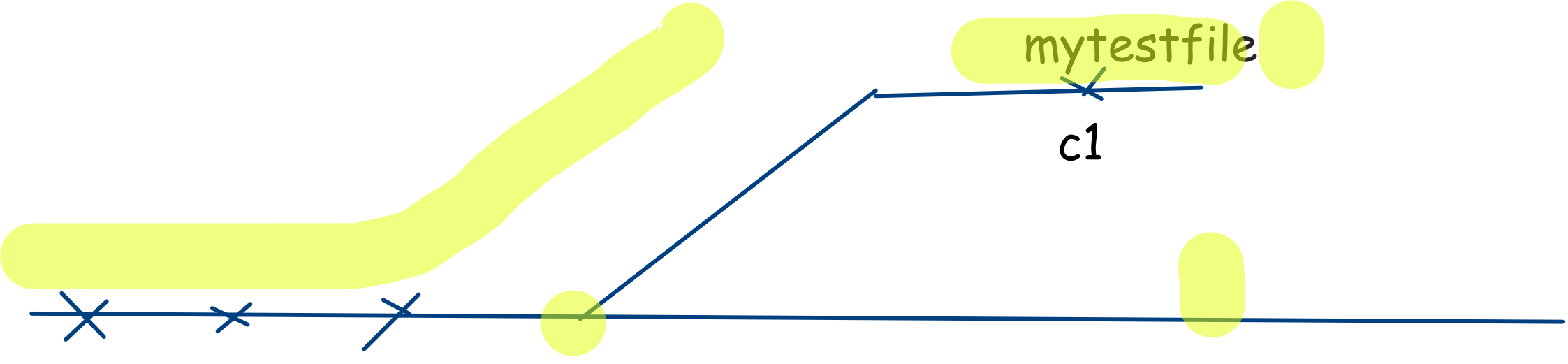
e.g git branch aboutusbranch
WHEN YOU PRACTICE CREATE BRANCH WITH DIFFERENT NAME

##
git branch
o/p:
* master <-----------------current branch
aboutusbranch

##
I want to move to aboutusbranch :

git checkout aboutusbranch

git branch
o/p:
master
*aboutusbranch



touch mytestfile
git add mytestfile
git commit -m"c1"

ls ## you will see with the list of files you will have mytestfile

git checkout master

ls ### you will see other list of files but mytestfile is missing
as branches are isolated from each other

```
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 mytestfile
akshat@Akshats-MacBook-Air apachewebsite % ls
buildspec.yml           images                  js
css                     index.html              mytestfile
deployment.yml          installapche.yml        service.yml
Dockerfile              inventory.ini           test
icon                    Jenkinsfile
akshat@Akshats-MacBook-Air apachewebsite % git branch
* aboutusbranch
  master
akshat@Akshats-MacBook-Air apachewebsite % git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
akshat@Akshats-MacBook-Air apachewebsite % ls
buildspec.yml           images                  js
css                     index.html              service.yml
deployment.yml          installapche.yml        test
Dockerfile              inventory.ini
icon                    Jenkinsfile
akshat@Akshats-MacBook-Air apachewebsite %
```

Keychain

This proves that branches are isolated from each other ..

Lets say at this point you reliase that the functionality is completed locally and you want to push the branch to github

So first of all you will request your project github admin to share the PAT with you

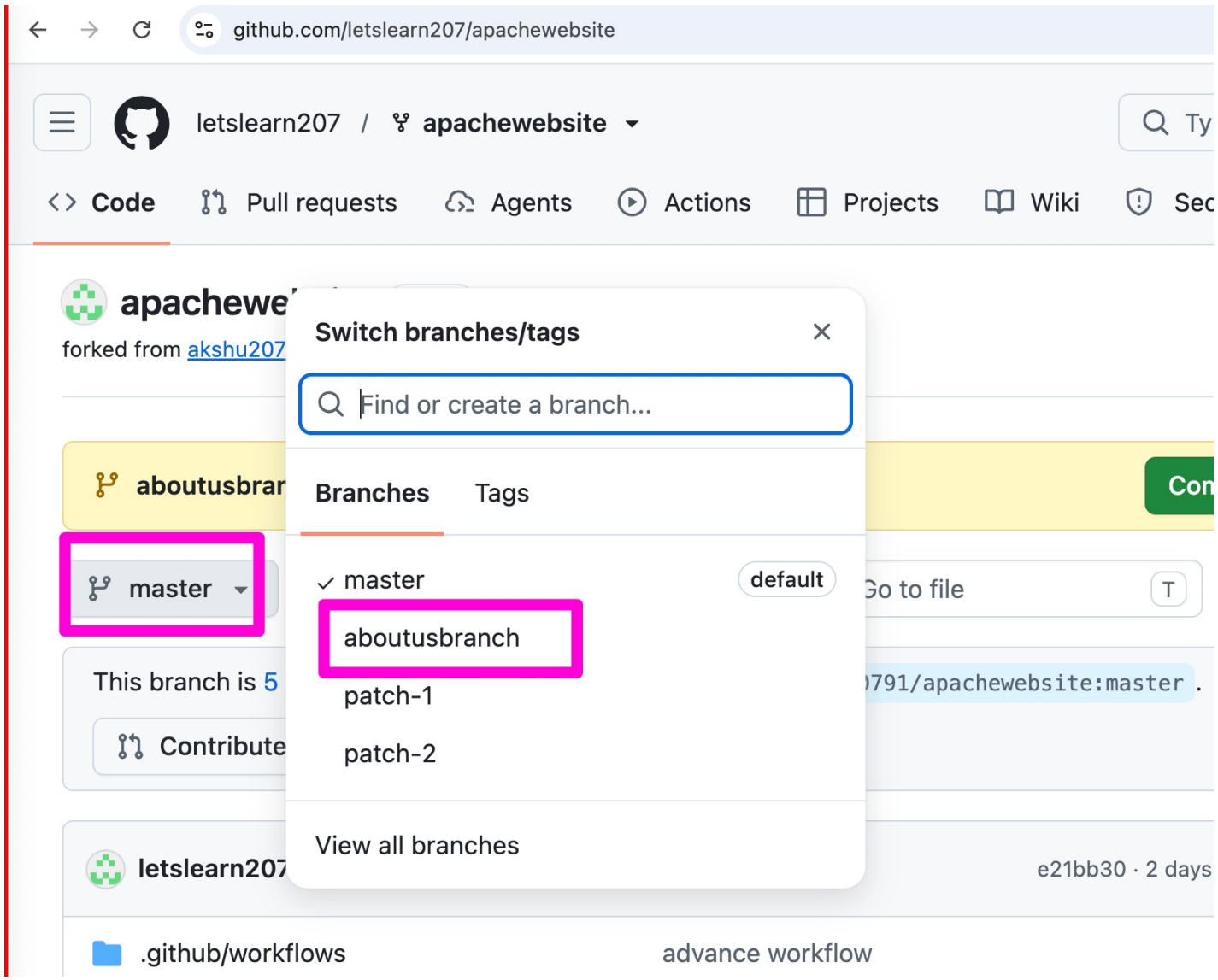
COPY THIS TOKEN:

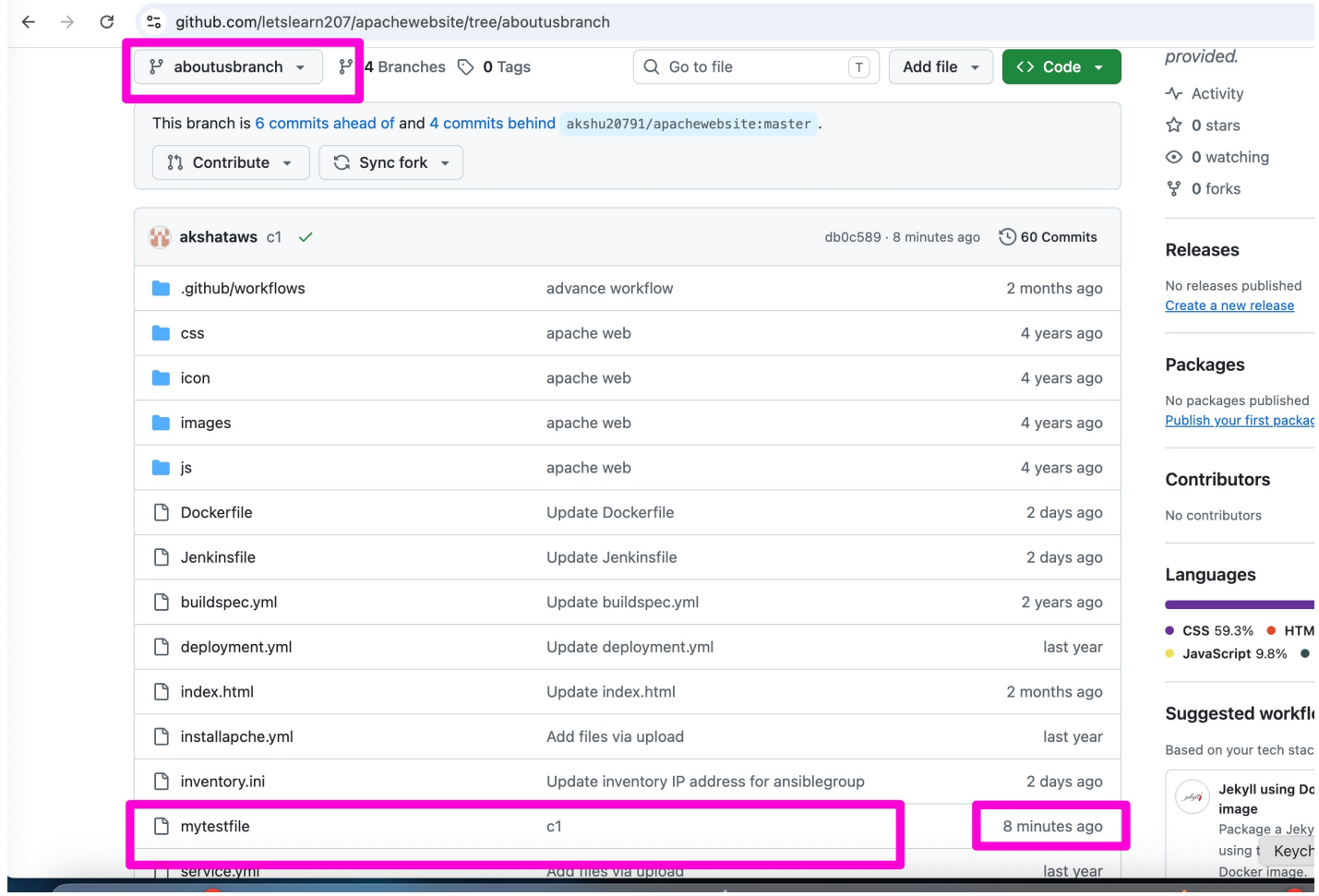
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

git push -u origin branchname

e.g git push -u origin aboutusbranch
pop will appear ...click on token and copy that token and continue

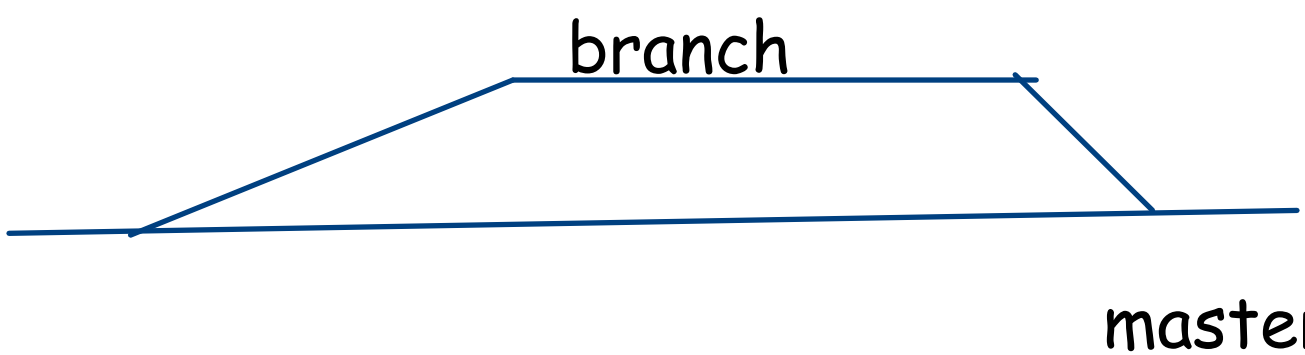
Go to github repo : https://github.com/letslearn207/apachewebsite




github.com/letslearn207/apachewebsite/tree/aboutusbranch
aboutusbranch
4 Branches
0 Tags
Go to file
T
Add file
Code
This branch is 6 commits ahead of and 4 commits behind akshu20791/apachewebsite:master.
Contribute
Sync fork
akshataws c1
db0c589 · 8 minutes ago
60 Commits
.github/workflows advance workflow 2 months ago
css apache web 4 years ago
icon apache web 4 years ago
images apache web 4 years ago
js apache web 4 years ago
Dockerfile Update Dockerfile 2 days ago
Jenkinsfile Update Jenkinsfile 2 days ago
buildspec.yml Update buildspec.yml 2 years ago
deployment.yml Update deployment.yml last year
index.html Update index.html 2 months ago
installapche.yml Add files via upload last year
inventory.ini Update inventory IP address for ansiblegroup 2 days ago
mytestfile c1 8 minutes ago
service.yml Add files via upload last year
provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first packa
Contributors
No contributors
Languages
CSS 59.3%
HTM
JavaScript 9.8%
Suggested workfl
Based on your tech stac
Jekyll using Do
image
Package a Jeky
using t
Docker image.
Keych

GIT MERGE :

you want to merge branch with another branch



git branch
(if you are not in master then do the checkout to master)
git checkout master
git merge branchname

```
branch 'aboutusbranch' set up to track 'origin/aboutusbranch'.
akshat@Akshats-MacBook-Air apachewebsite % git branch
  aboutusbranch
* master
akshat@Akshats-MacBook-Air apachewebsite % git merge aboutusbranch
Updating e21bb30..db0c589
Fast-forward
 mytestfile | 0
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 mytestfile
akshat@Akshats-MacBook-Air apachewebsite % ls
buildspec.yml           images                  js
css                     index.html              mytestfile
deployment.yml          installapche.yml        service.yml
Dockerfile              inventory.ini           test
icon                    Jenkinsfile
akshat@Akshats-MacBook-Air apachewebsite %
```

file we created in aboutusbran
is there is master